SILAS BAMBIL

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EDUCATION

Georgia State University

United States | MS Mathematics, concentration in Statistics | August 2026-present

Georgia State University

United States | MS Quantitative Risk Analysis and Management | May 2025

Kwame Nkrumah University of Science and Technology

Ghana | BSc Actuarial Science | July 2020

RESEARCH INTERESTS

Statistical model updating under distribution shift; selection regret and predictive validation; calibration and credit risk; extreme value theory; expected shortfall and bounded-layer diagnostics; simulation-based inference.

CURRENT RESEARCH MANUSCRIPTS

Tail Model Diagnostics Using Expected Shortfall and Bounded Loss Layers

with an Application to Fire Losses | 2026 | Manuscript in preparation

With Shudeshna Nandy, Clement Ampong, Prakash Karki, and Chi-Kuang Yeh. Develops bounded-layer comparisons and inferential procedures for assessing discrepancies between fitted tail models and empirical loss distributions. Uses U.S. structure-fire losses from NFIRS (2016-2025); distinguishes discrepancy detection from localization.

Selection Regret in Credit-Risk Model Updating Under Distribution Shift:

The Role of Recent Labeled Information | 2026 | Independent manuscript in preparation

Studies whether selecting the best fitted updating policy more frequently leads to smaller predictive regret. Simulation experiments across 13 scenarios compare adaptation sample sizes and separate oracle agreement from the predictive cost of selection errors; includes a historical mortgage-data application.

Selecting and Estimating Credit-Risk Model Updates Under Distribution Shift

2026 | Manuscript in preparation | With Shudeshna Nandy

Examines allocation of recent labeled observations between choosing an updating policy and estimating the selected model. Compares retention, recalibration, unrestricted and regularized refitting through simulation and historical mortgage cohorts, with independent predictive evaluation.

EARLIER ACADEMIC RESEARCH

Graduate Admissions Predictive Analysis | Georgia State University

United States | 2024

Prepared applicant data, developed candidate predictors, and contributed to evaluation of graduate-enrollment prediction models.

Application of Extreme Value Theory to Fire Insurance | Undergraduate Thesis

Kwame Nkrumah University of Science and Technology | Ghana | 2020

Examined extreme value methods for modeling severe fire-insurance losses.

ACADEMIC APPOINTMENTS

Graduate Assistant | Georgia State University

United States | August 2023-June 2025

- Supported predictive enrollment analysis and faculty risk-analysis projects through statistical analysis, research assistance, and data management.

TEACHING EXPERIENCE

Graduate Lab Assistant | Georgia State University

United States | August 2026-present

- Support students in College Algebra (MATH 1111) and Support for College Algebra (MATH 0999), with emphasis on mathematical reasoning and problem solving.

AP Mathematics Teacher | Osceola School for the Arts

United States | January-June 2026

- Taught AP Calculus, AP Precalculus, and AP Statistics; prepared lessons and assessments and provided individual support. Reported AP examination pass rate: 70%.

PROFESSIONAL EXPERIENCE

Finance Associate | OPH CHEM CONSULT

Ghana | May 2022-August 2023

- Conducted financial research on misappropriation risks and community financial challenges, contributing to analytical reports and briefs.

Research Assistant | Ghana Statistical Service

Ghana | September 2020-August 2021

- Contributed statistical analysis and data-management support to population and economic studies.

TEACHING AND RESEARCH SKILLS

Lesson planning, classroom management, assessment design, quantitative instruction; research design and methodological frameworks; simulation design, predictive validation, sensitivity analysis, bootstrap inference, and academic writing.

TECHNICAL SKILLS

Programming: R, Python, Microsoft Excel and Microsoft Office. Statistical methods: regression, Monte Carlo simulation, model calibration, bootstrap methods, data management, and visualization. Risk methods: Peaks-Over-Threshold, Generalized Pareto models, Value-at-Risk, Expected Shortfall, and credit-risk modeling.

AWARDS

- Graduate Merit Scholarship, Department of Mathematics and Statistics, Georgia State University | August 2026.
- Graduate Merit Scholarship, Robinson College of Business, Georgia State University | August 2023-May 2025.

FINANCIAL-INDUSTRY EXAMINATIONS

- FINRA Securities Industry Essentials (SIE) Examination | January 2025.
- FINRA Series 63 - Uniform Securities Agent State Law Examination | August 2025.